



Economic Analysis of Financial Markets

Fixed Income Securities

Test 1: 27th November 2023

75 Marks

135 Minutes

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY:

1. PLEASE DO NOT TURN THIS PAGE OVER UNTIL INSTRUCTED TO DO SO.
2. Ensure that your name and student number appears on the cover pages of ALL answer booklets.
3. All cell phones must be switched off for the duration of the examination and placed with your other possessions out of sight underneath your chair.
4. You must remain seated until every paper has been collected.
5. The invigilators will answer no questions relating to the paper during the test. Make whatever assumptions you deem appropriate and clearly state these in your answer.
6. You can answer all questions in one book.
7. Show all your workings. No marks will be awarded for answers that are not backed up with appropriate workings
8. Write in full and avoid using symbols. Acronyms should be defined on each question.

This test paper consists of 7 pages including the cover page. The formula sheet and calendars are found at the end of the paper

QUESTION 1

[25 Marks]

The image below displays a snapshot of a bond issued by the South African government.

As of 24		3.595 / 2.551		Source B	
Actions		Settings		Page 1/13 Security Description: Bond	
94 Notes		95 Buy		96 Sell	
15 Bond Description		20 Issuer Description			
Pages		Issuer Information		Identifiers	
11 Bond Info		Name REPUBLIC OF SOUTH AFRICA		FIGI BBG00000L512	
12 Addtl Info		Industry Treasury (BCLASS)		ISIN ZAG000018003	
13 Reg/Tax		Security Information		ID Number EC3963484	
14 Covenants		Mkt Iss DOMESTIC		Bond Ratings	
15 Guarantors		Ctry/Reg ZA		Moody's Ba2	
16 Bond Ratings		Currency ZAR		S&P BB	
17 Identifiers		Rank Sr Unsecured		Fitch BB-	
18 Exchanges		Series R197		Composite BB-	
19 Inv Parties		Coupon 5.500000		Issuance & Trading	
20 Fees, Restrict		Type Fixed		Amt Issued/Outstanding	
21 Schedules		Cpn Freq S/A		ZAR 33,757,517.48 (M) /	
22 Coupons		Day Cnt ACT/365		ZAR 27,004,517.48 (M)	
23 Impact		Maturity 12/07/2023		Min Piece/Increment	
Quick Links		BULLET		1.00/ 1.00	
33 ALLQ Pricing		Iss Sprd		Par Amount 1.00	
34 QRD Qt Recap		Calc Type		Book Runner	
35 TDH Trade Hist		Auction Date 05/09/2001		Exchange JSE Contrib Prx	
36 CACS Corp Action		1st Coupon Date 06/07/2001			
37 CF Filings		Exchange Notice Date 11/16/2023			
38 CN Sec News		Exchange Expiration Date 11/16/2023			
39 HDS Holders					
40 Send Bond					

Additional information:

The “books closed dates” for this bond are 27th November and 28th May.

Required

1. Calculate the current yield for this bond. Assume you placed an order to buy the bond today. Work with a par value of R10,000. [19]
2. What is the relationship between the current yield and yield-to-maturity for this bond? Provide an explanation to justify the observed relationship. [4]
3. Specify one difference between the yield-to-maturity and realised compound yield. [2]

QUESTION 2**[16 Marks]**

1. You are a bond trader managing a portfolio primarily composed of short-term bonds (with maturities of 1-year and less). You are anticipating a significant increase in the repurchase rate (repo rate) next week. How would you restructure your portfolio, now (today), to maximise returns from your forecast. Support your answer by explaining how your portfolio position generates positive returns if your forecast materialises. **[4]**

2. A surprise reduction in the repurchase rate (repo rate) can lead to a decline in bond yields. Explain the mechanism that triggers this shift, detailing the process from the repo rate cut to the subsequent decrease in bond yields. **[3]**

3. The South African Reserve Bank convened its Monetary Policy meeting on Thursday, November 23, 2023.
 - a. What decision was made regarding the repurchase rate? Provide one potential rationale for the choice they made. **[3]**

 - b. The upcoming Monetary Policy Committee meeting is scheduled for January 25, 2023. Based on your interpretation of the current monetary policy statement, what is the next probable course of action regarding the repurchase rate, and could you provide a justification for your answer? **[3]**

4. What is your projection (or forecast) of future interest rates over the next five-years (increase/decrease or remain constant). Provide a justification for your forecast? **[3]**

QUESTION 3**[4 Marks]**

The table below displays three bonds along with their respective features. Without performing any calculations, identify which of these three bonds is inaccurately quoted. Provide a justification for your response. **[4]**

Bond Name	Coupon rate	Yield-to-maturity	Face Value	Price
Bond A	8%	3%	R11,000	R7,785.61
Bond B	12%	8%	R18,000	R22,831.26
Bond C	10%	12%	R15,000	R13,304.93

QUESTION 4**[10 Marks]**

1. Explain why you are more likely to misprice bonds when using the traditional method for bond valuation. Explain how the arbitrage-free valuation addresses this shortcoming of the traditional method? **[5]**
2. You strongly believe that a recession is coming soon. You want to create a portfolio, today, which maximises returns during the recession. Considering this, do you maximise returns by buying short- or long-term bonds? Provide two reasons to justify your answer. **[5]**

QUESTION 5**[20 Marks]**

The table provided below displays three government bonds, each with a face value of R1,000.

Maturity (year)	Maturity date	Coupon rate	Price (R)
1	27 th November 2024	0%	R934.58
2	27 th November 2025	10%	R1,069.58
3	27 th November 2026	12%	R1,234.78

Required

1. Assuming the pure expectations theory holds, calculate the implied one-year forward interest rate that is expected to prevail on the 27th November 2024. **[9]**
2. State one reason why it is important to know forward interest rates. **[2]**
3. Based on the information presented in the above table, calculate the arbitrage-free value of a two-year ShonCom bond with a face value is R1,000 and a coupon rate of 12%. **[4]**
4. Given the arbitrage free value from the previous question, describe a profitable trade that you can execute if the ShonCom bond is trading at a market price of R850. Specify the trading profit (assuming no transaction costs). **[5]**

EAFM-Formula Sheet 1

$D = \frac{\left(\sum_{t=1}^T PV(CF_t) \times t \right)}{P_0}$
$D_{perpetuity} = \frac{(1+y)}{y}$
$D_{annuity} = \frac{1+y}{y} - \frac{T}{(1+y)^T - 1}$
$D = \frac{1+y}{y} - \frac{(1+y) + T(c-y)}{c[(1+y)^T - 1] + y}$
$D_{parbond} = \frac{1+y}{y} \left[1 - \frac{1}{(1+y)^T} \right]$
$FVIFA_{ordinary} = \frac{[(1+r)^n - 1]}{r}$
$FVIFA_{due} = \frac{[(1+r)^n - 1] \times (1+r)}{r}$
$PVIFA_{ordinary} = \frac{\left[1 - \frac{1}{(1+r)^n} \right]}{r}$
$PVIFA_{due} = \left[\frac{\left[1 - \frac{1}{(1+r)^n} \right]}{r} \right] \times (1+r)$
$Convexity = \frac{(PV_-) + (PV_+) - 2 \times (PV_0)}{(\Delta yield)^2 \times PV_0}$
$D^* = \frac{(PV_-) - (PV_+)}{2 \times (\Delta Yield) \times (PV_0)}$
$i_{1,H} = i_{1,L} e^{2\sigma} \text{ — i tree}$

Calendar for Year 2023 (South Africa)

January Su Mo Tu We Th Fr Sa 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 7:0 15:0 21:0 28:0	February Su Mo Tu We Th Fr Sa 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 5:0 13:0 20:0 27:0	March Su Mo Tu We Th Fr Sa 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 7:0 15:0 21:0 29:0
April Su Mo Tu We Th Fr Sa 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 6:0 13:0 20:0 27:0	May Su Mo Tu We Th Fr Sa 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 5:0 12:0 19:0 27:0	June Su Mo Tu We Th Fr Sa 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 4:0 10:0 18:0 26:0
July Su Mo Tu We Th Fr Sa 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 3:0 10:0 17:0 26:0	August Su Mo Tu We Th Fr Sa 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 1:0 8:0 16:0 24:0 31:0	September Su Mo Tu We Th Fr Sa 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 7:0 15:0 22:0 29:0
October Su Mo Tu We Th Fr Sa 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 6:0 14:0 22:0 28:0	November Su Mo Tu We Th Fr Sa 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 5:0 13:0 20:0 27:0	December Su Mo Tu We Th Fr Sa 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 5:0 13:0 19:0 27:0

23

92

68
= 183

Holidays:

1 Jan New Year's Day	27 Apr Freedom Day	25 Sep 'Heritage Day' day off
2 Jan 'New Year's Day' day off	1 May Workers' Day	16 Dec Day of Reconciliation
21 Mar Human Rights Day	16 Jun Youth Day	25 Dec Christmas Day
7 Apr Good Friday	9 Aug National Women's Day	26 Dec Day of Goodwill
10 Apr Family Day	24 Sep Heritage Day	

As of 24 3.595 / 2.551 -->-- Source B

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Issuer Information

Name REPUBLIC OF SOUTH AFRICA

Industry Treasury (BCCLASS)

Security Information

Mkt Iss DOMESTIC

Ctry/Reg ZA Currency ZAR

Rank Sr Unsecured Series R197

Coupon 5.500000 Type Fixed

Cpn Freq S/A

Day Cnt ACT/365 Iss Price 100.0000

Maturity 12/07/2023

BULLET

Iss Sprd

Calc Type

Auction Date 05/09/2001

1st Coupon Date 06/07/2001

Exchange Notice Date 11/16/2023

Exchange Expiration Date 11/16/2023

Identifiers

FIGI BBG000000L512

ISIN ZAG000018003

ID Number EC3963484

Bond Ratings

Moody's Ba2

S&P BB

Fitch BB-

Composite BB-

Issuance & Trading

Amt Issued/Outstanding

ZAR 33,757,517.48 (M) /

ZAR 27,004,517.48 (M)

Min Piece/Increment

1.00 / 1.00

Par Amount 1.00

Book Runner

Exchange JSE Contrib Prx